

Modeling viral kinetics

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Daily longitudinal mechanistic models of SARS-CoV-2 infection reveals substantial heterogeneity in infectiousness

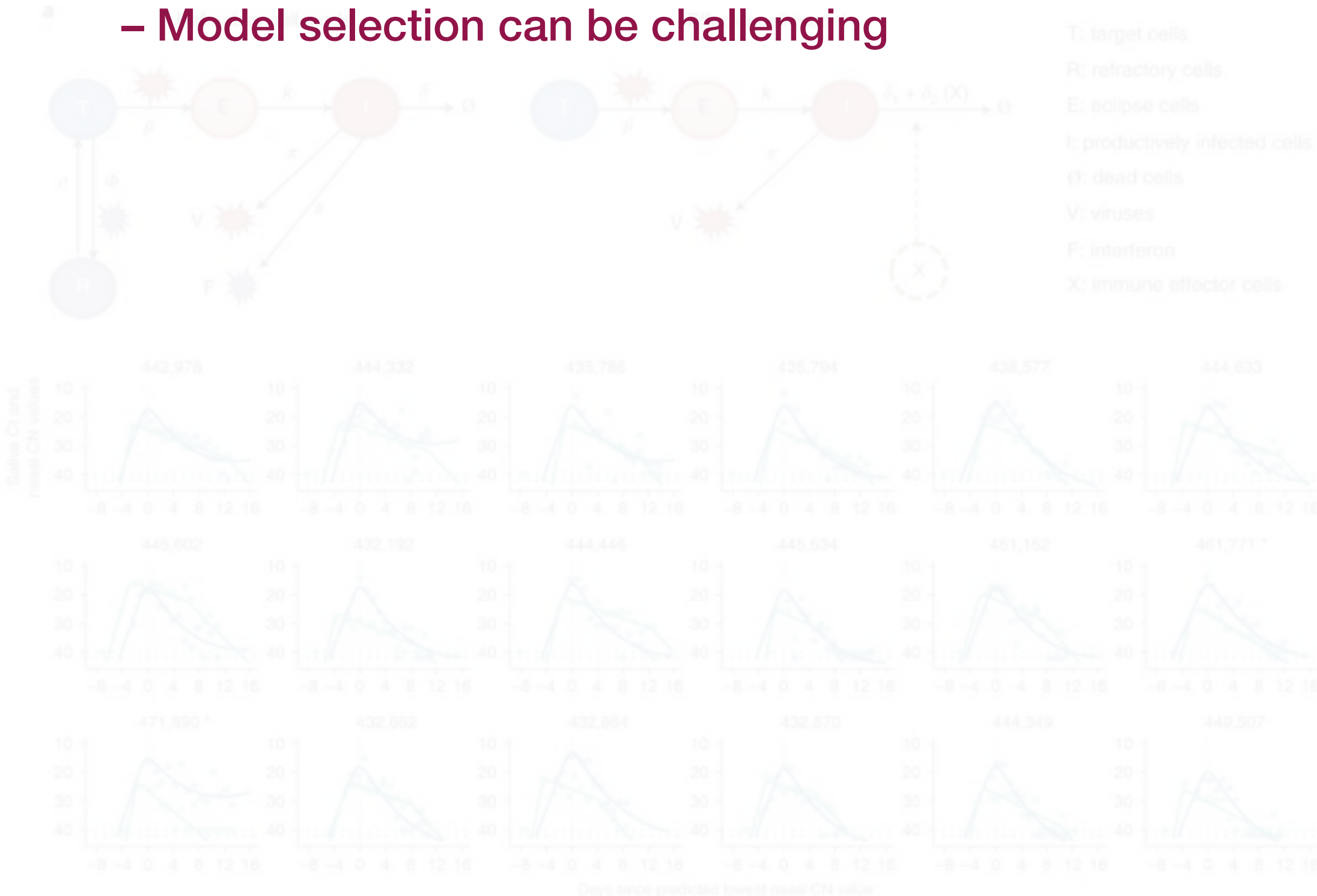
Ruian Ke, Pamela P. Martinez, Rebecca L. Smith, Laura L. Gibson, Agha Mirza, Madison Conte, Nicholas

+ Give insights about immunological mechanisms

+ Can be used to predict viral kinetic features (e.g., rebound)

– Can be challenging to fit/specify the parameters

– Model selection can be challenging



Viral kinetics Phenomenological models

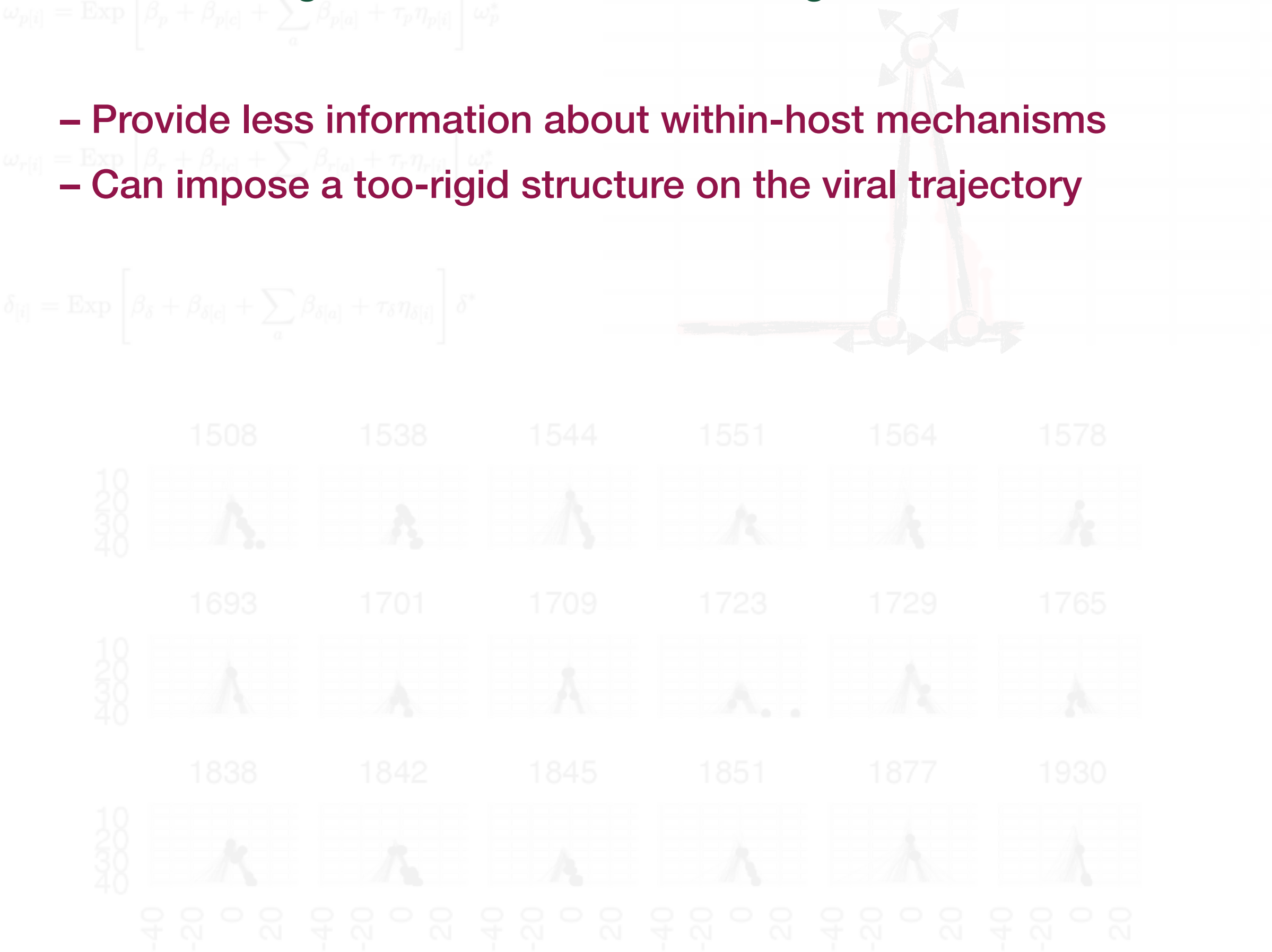
Stephen M. Kissler, James A. Hay, Joseph R. Fauver, Christina Mack, Caroline G. Tai, Deverick J.

+ Provide epidemiologically relevant information with robust uncertainty measures

+ Quick/straightforward to fit, even to large datasets

– Provide less information about within-host mechanisms

– Can impose a too-rigid structure on the viral trajectory



Some history

THE GROWTH OF BACTERIOPHAGE

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(Accepted for publication, September 7, 1938)

INTRODUCTION

Certain large protein molecules (viruses) possess the property of multiplying within living organisms. This process, which is at once so foreign to chemistry and so fundamental to biology, is exemplified in the multiplication of bacteriophage in the presence of susceptible bacteria.

Bacteriophage offers a number of advantages for the study of the multiplication process not available with viruses which multiply at the expense of more complex hosts. It can be stored indefinitely in the absence of a host without deterioration. Its concentration can be determined with fair accuracy by several methods, and even the individual particles can be counted by d'Herelle's method. It can be concentrated, purified, and generally handled like nucleoprotein, to which class of substances it apparently belongs (Schlesinger (1) and Northrop (2)). The host organism is easy to culture and in some cases can be grown in purely synthetic media, thus the conditions of growth of the host and of the phage can be controlled and varied in a quantitative and chemically well defined way.

